Zixuan Guo

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RESEARCH INTERESTS

Music & Audio Representation Learning · Machine Music Understanding · Music Information Retrieval

EDUCATION

New York University — New York, NY
Master of Music (M.M.) in Music Technology — *Sep 2024 – May 2026*
GPA: 3.98/4.0

Jilin University — Jilin, China
Bachelor of Science (B.S.) in Computer Science and Technology — *Sep 2019 – Jun 2024*

Berklee College of Music — Boston, MA
Five-Week Summer Performance Program (Woodwind Performance) — *Jul 2023 – Aug 2023*

RESEARCH EXPERIENCE

Master's Thesis — Controllable Symbolic Music Generation for Human-AI Music Creation *2025–2026*
New York University Code

- Studied symbolic music representations and musical structure to enable fine-grained, iterative control in DAW-based creative workflows.
- Developed a hybrid framework combining MuseCoco generation, deterministic symbolic constraints, and AR-VAE latent-space editing across New, Continue, and Transformation modes.
- Built a JUCE DAW plugin with an inference backend for non-destructive generation and refinement.
- Evaluated controllability through ablations, objective constraints, and a 7-participant user study, improving New-mode constraint success from 48% to 76% (+28 pp).

Noise-Robust Musical Instrument Recognition *2025*
New York University Code

- Investigated representation robustness under acoustic distribution shift, evaluating LAION-CLAP embeddings across progressively degraded SNR conditions.
- Constructed controlled noisy variants of Medley-solos-DB using MUSAN noise and developed noise-aware curriculum training for TCN and lightweight Transformer classifiers.
- Achieved 95.93% clean accuracy and 81.28% accuracy at -5 dB SNR; introduced pure-noise negatives and a confidence-triggered harmonic fallback for explicit rejection under extreme noise.

Research Assistant — DeepSqueak *May 2025 – Nov 2025*
University of Washington, Seattle, WA Code

- Improved robustness of a legacy spectrogram-based acoustic event detection pipeline under low-SNR bioacoustic conditions.
- Replaced YOLOv2 with anchor-free YOLOX and developed multi-channel spectrogram preprocessing, dynamic noise augmentation, and hard-negative mining.
- Conducted failure-case analysis and targeted data refinement to improve event localization and reduce manual bounding-box correction.

Low-Label Acoustic Scene Classification under DCASE Resource Constraints *2026*
New York University Code

- Studied low-label acoustic representation learning under DCASE 2024 Task 1 resource constraints (< 128 KB parameters; < 30 MMAC/clip).
- Reproduced the ConvMixer+RSC baseline and built a deterministic evaluation pipeline covering fixed splits, augmentation, device/class metrics, checkpointing, and compute profiling.
- Found that EfficientLEAF parameters remained close to their mel initialization under 5% supervision and produced no measurable gain over the baseline frontend.

- Evaluated Efficient Channel Attention as a lightweight alternative, improving macro accuracy while reducing compute from ∼29.4 to ∼26.5 MMACs.

Bachelor's Thesis — Domain-Adversarial Transfer Learning *2024*
Jilin University Code

- Studied unsupervised domain adaptation by introducing class-matched Mixup/CutMixup intermediate domains within a DANN framework.
- Developed a staged source → intermediate → target adaptation pipeline and evaluated it across Office-31, ImageCLEF, and OfficeHome, improving ImageCLEF accuracy from 76.84% to 95.00%.

PUBLICATIONS & PREPRINTS

- **Guo, Z.**, Fang, Q., Yang, J.-S., & Rhee, M. (2026). *Closed-Loop Audio Neuroregulation for ADHD: Graph-Based Attention Decoding and Rhythm Entrainment*. *IEEE Access*. doi:10.1109/ACCESS.2026.3731552.
- **Guo, Z.**, Zhong, J., & Chen, S. (2025). *Noise-Robust Instrument Classification under Real-World SNR Conditions*. *SSRN Electronic Journal* (Preprint).
- Mo, T., Zhang, C., Zou, J., **Guo, Z.**, & Rhee, M. (2026). *Self-Evolving AI Agents with Dual Memory for Automated Software Testing and Bug Localization*. *IEEE Access*. doi:10.1109/ACCESS.2026.3713401.
- Zhang, F., **Guo, Z.**, Ding, J., Yang, J., & Liu, W. (2026). *Adaptive Sensor Fusion for Robust Perception in Dense Fog: A Gated Vision and LiDAR Integration Framework*. *Sensors*. doi:10.3390/s26123728.

INDUSTRY EXPERIENCE

Machine Learning Intern — APEXUS-TECH Remote
Jul 2026 – Aug 2026

- Built and evaluated a retrieval-augmented ML research system using hybrid retrieval, rank fusion, structured generation, and provenance validation.
- Developed time-dependent ML evaluation pipelines with expanding-window validation and explicit leakage checks.

Data Mining & Analytics Intern — Hazel Investment Shenzhen, China
Jan 2024 – Feb 2024

- Conducted Python-based financial data analysis, visualization, and automated reporting.

ACADEMIC SERVICE

- Invited Reviewer, Scientific Reports (Nature Portfolio), 2026
- Invited Reviewer, Discover Artificial Intelligence (Springer Nature), 2026

HONORS & AWARDS

- **Academic Achievement Recognition**, Graduate Music Technology Program, NYU Steinhardt *May 2026*
- **Third-Class Scholarship**, Department of Computer Science and Technology, Jilin University *Sep 2020*

TECHNICAL SKILLS

Audio & Music AI: Music Information Retrieval, Machine Listening, Symbolic Music Generation, Audio DSP, Acoustic Event Detection
Machine Learning: PyTorch, Transformers, VAEs, Representation Learning, Domain Adaptation
Audio Software & Creative Computing: JUCE, Max/MSP, SuperCollider, Ableton Live, Pro Tools
Programming: Python, C/C++, MATLAB, SQL
Systems & Tools: Linux, Git, Docker, REST APIs